

GridForge Plant Operations Platform is the utility's control-adjacent platform for remote maintenance, supplier coordination, telemetry brokering, and predictive analytics. The environment is segmented into an external access zone, a plant DMZ, and the OT core where field gateways, command data, and the historian live. What makes this system difficult is that it mixes remote human access, high-volume telemetry, and maintenance automation with assets that can affect plant safety.

Suppliers and remote maintenance vendors authenticate through a VPN Gateway and land on an OT Jump Host. Approved sessions are brokered to the Maintenance Orchestrator, which can dispatch jobs to Field Gateway nodes and request data from the Plant Historian. Telemetry from plant equipment streams through the Telemetry Broker, while predictive analytics run in the DMZ and consume selected historian or live data.

Engineers are especially worried about remote maintenance abuse, manipulation of command or recipe data, denial of view into plant conditions, and lateral movement from contractor access into the OT core. Analytics are valuable, but they are not supposed to become a path for control commands into the plant.